PEATLAND FIRE

under the 2023 to 2025 El Nino

Where Indonesian peat burns, why it burns,
and where the smoke travels

A reproducible analysis built entirely from open satellite and climate data

Five districts across Sumatra and Kalimantan

Why Peat Fires Matter

The scale and the challenge

- **Worst since 2019**

  The 2023 El Nino drove Indonesia's most severe fire season in years.

- **Hard to put out**

  Peat smoulders underground, resists suppression, and burns for weeks.

- **Carbon and haze**

  Peat fires release vast carbon and send haze across national borders.

- **The core question**

  With limited budgets, where should agencies act, and why there?

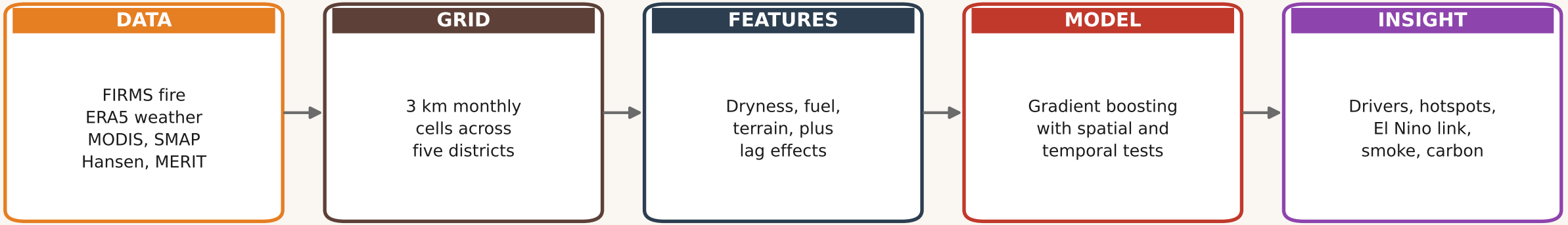
Data and Methods

How the analysis was built

- Built entirely from open satellite and climate data, with no proprietary inputs.
- Sources include NASA FIRMS fire, ERA5 Land weather, MODIS, SMAP soil moisture, and Hansen forest cover.
- A monthly fire susceptibility model was trained across five high risk districts for 2023 to 2025.
- It was validated both across years and across districts the model had never seen.
- This is a screening and attribution study, not an operational fire spread forecaster.

The Analytical Pipeline

From raw satellite data to a risk map



Validation

Temporal hold out across years, plus leave one district out spatial testing, confirms the model learns a transferable mechanism rather than memorising places.

Scale

143,478 monthly cells, 1,107 fire events, dry season months June to November.

Key Findings at a Glance

Every figure traces directly to the data

Ogan Komering Ilir

Highest risk district

r = 0.684

El Nino linkage to fire

0.708

Spatial validation score

239

Intensifying hotspot cells

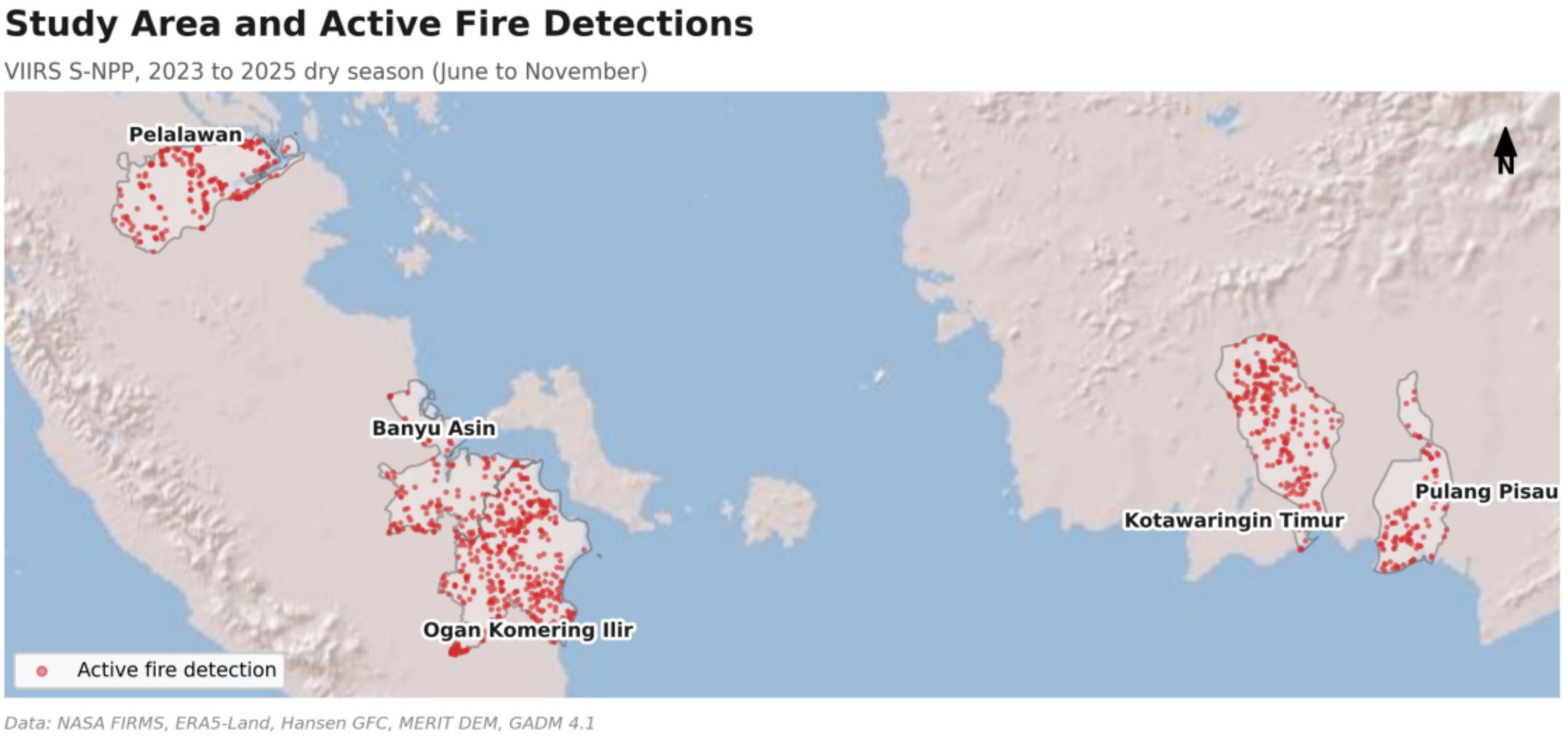
70.2-154.0 Mt

Estimated CO2, indicative

0.20 to 0.05

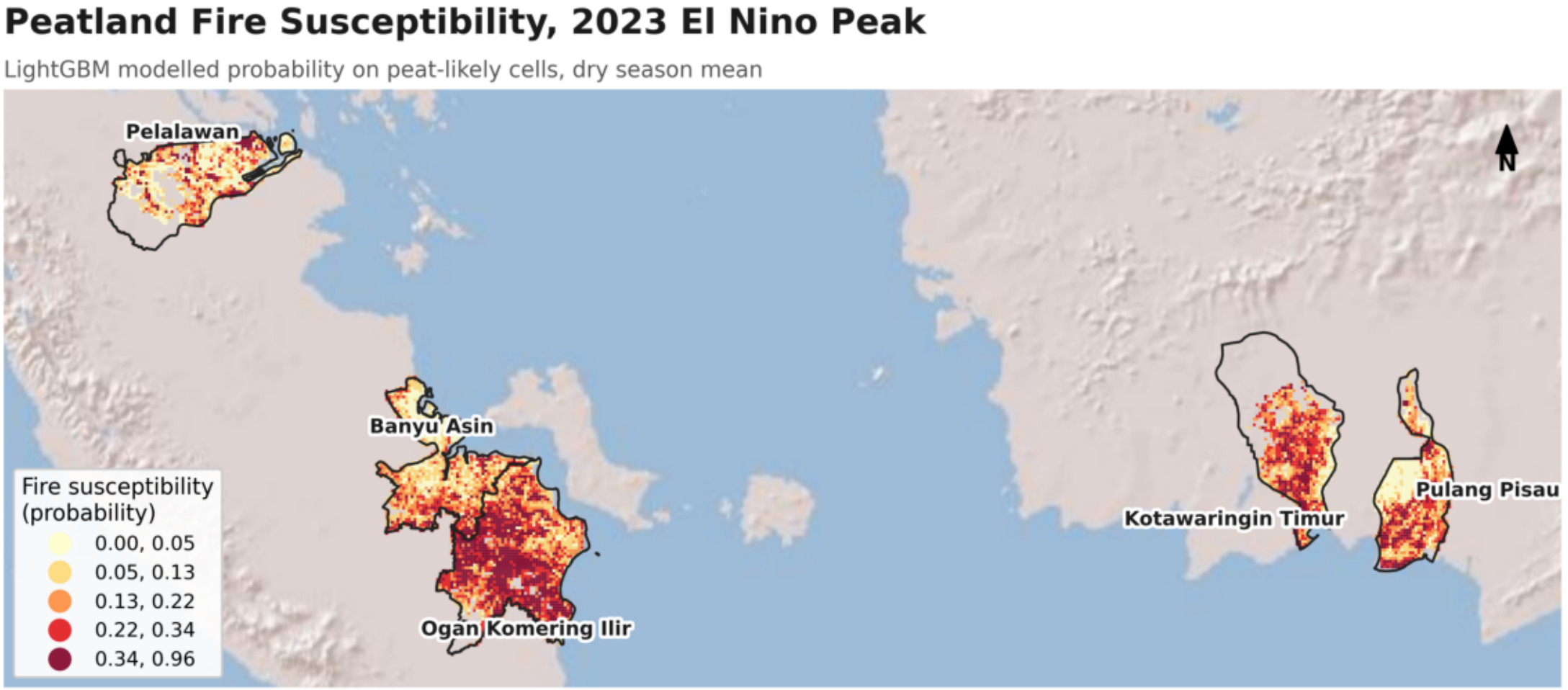
Risk fell, 2023 to 2025

Study Area and Fire Detections



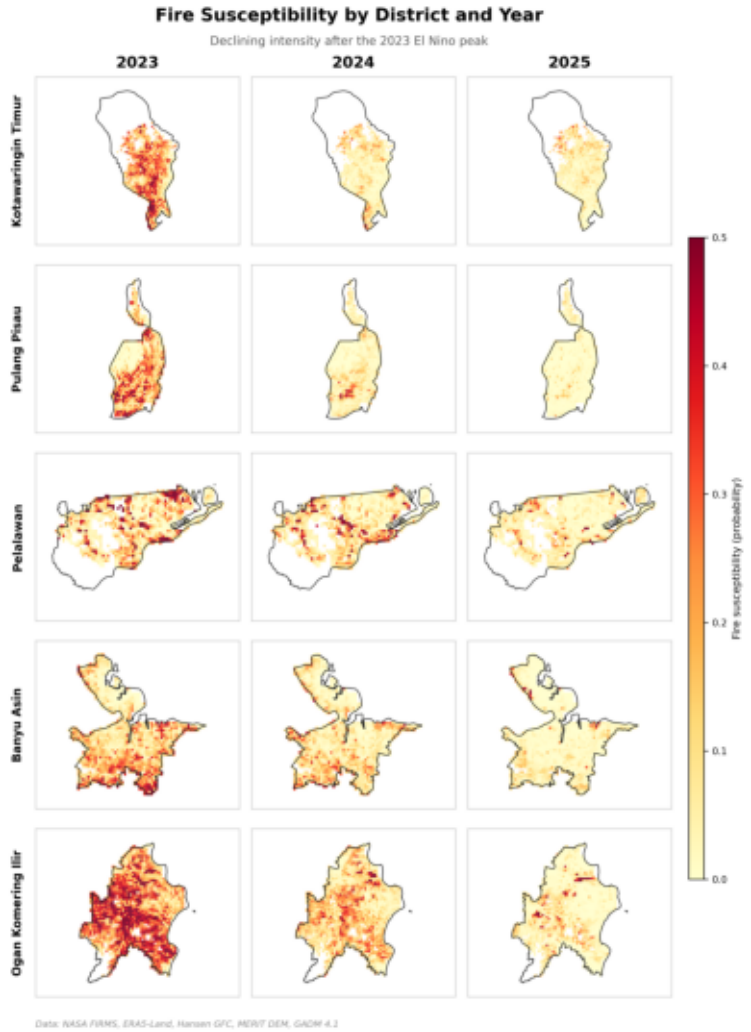
Every active fire detection across 2023 to 2025. Fire concentrates in the South Sumatra and Central Kalimantan peatlands.

Fire Susceptibility at the 2023 Peak



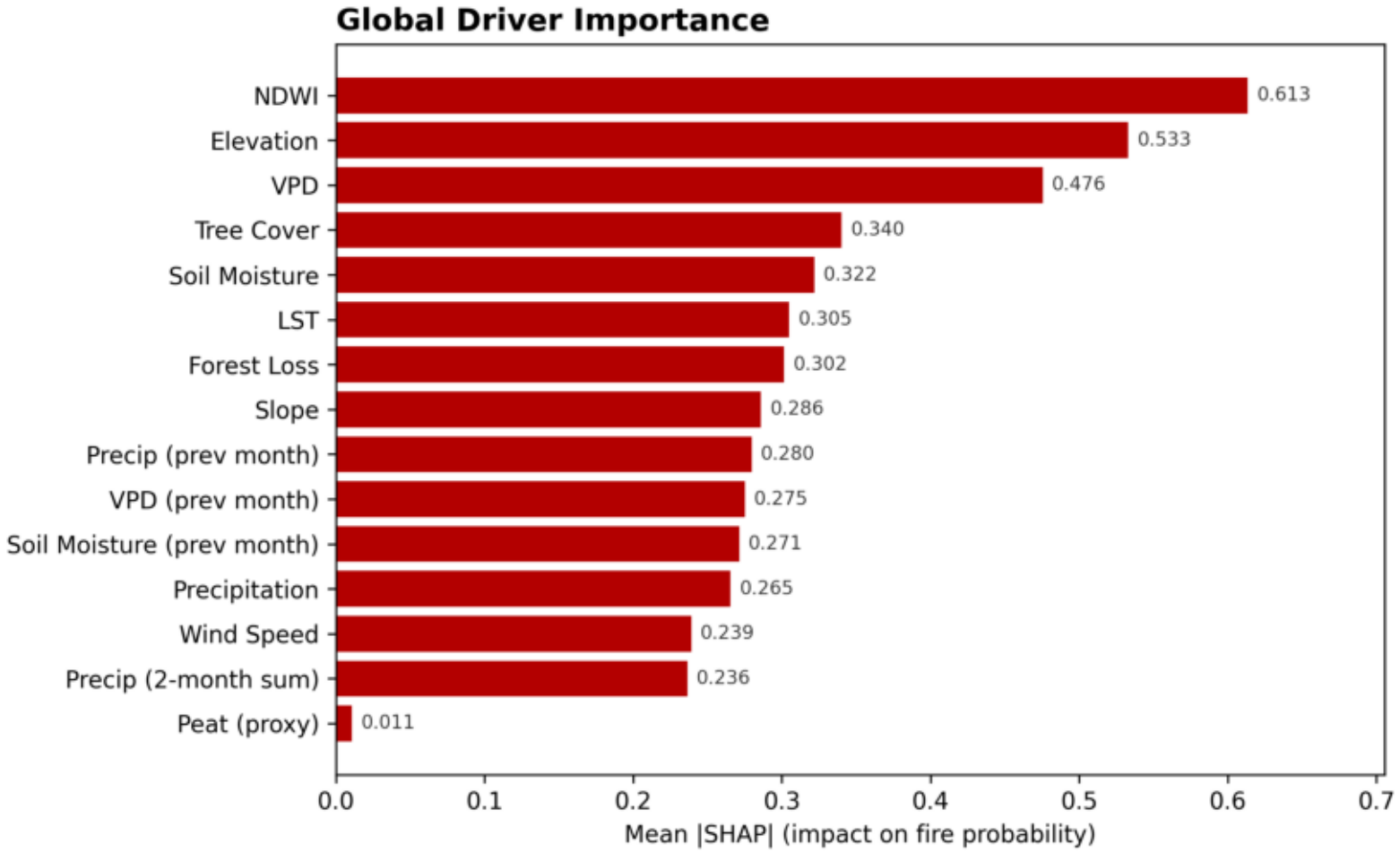
Modelled fire probability at the 2023 El Nino peak. Ogan Komering Ilir carries the highest risk.

Susceptibility by District and Year



Districts as rows, years as columns. Intensity fades from 2023 to 2025 as El Nino weakens.

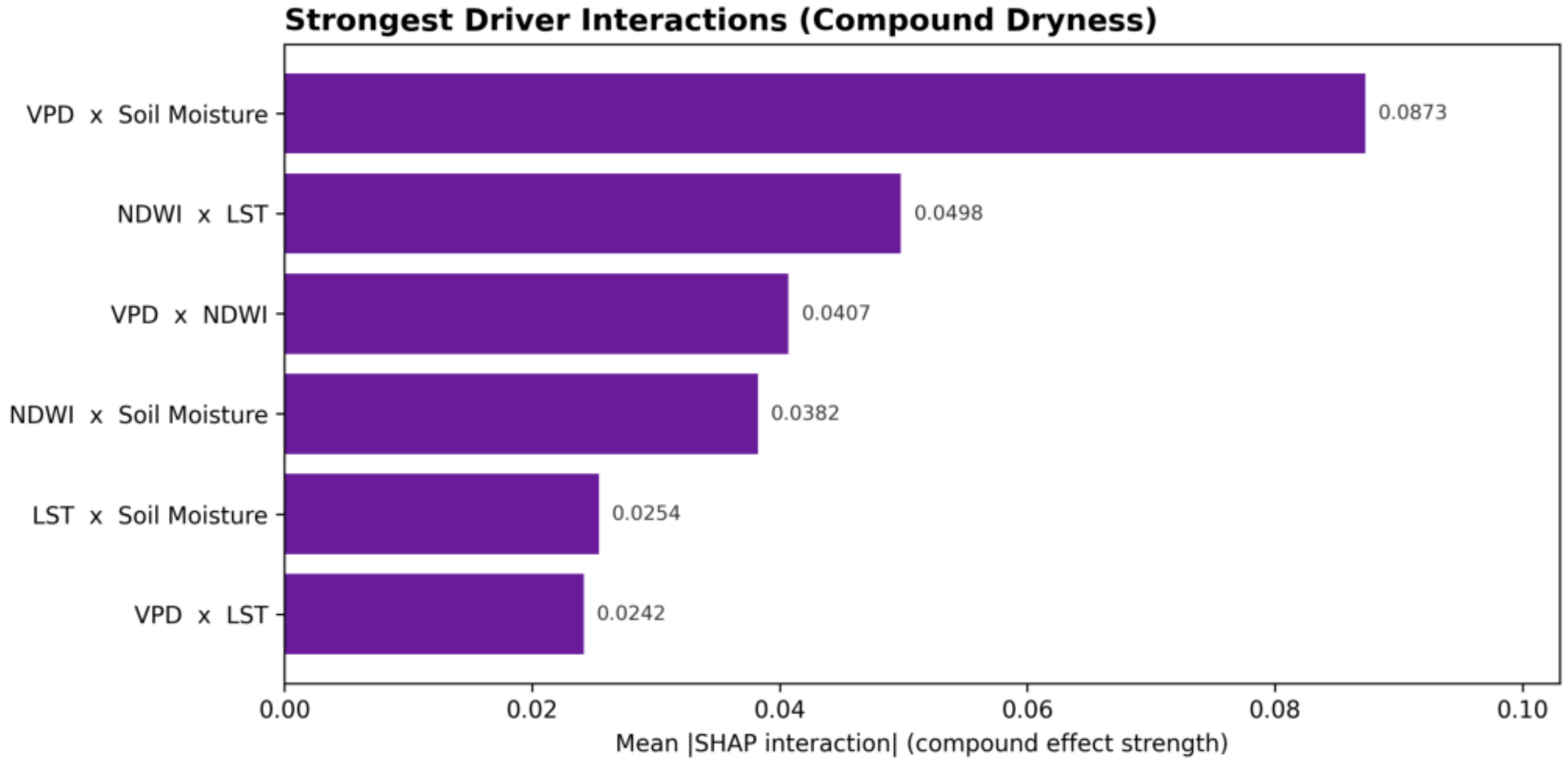
What Drives the Fire



NDWI: Normalized Difference Water Index | VPD: Vapour Pressure Deficit | LST: Land Surface Temperature

Dynamic dryness indicators, NDWI, VPD, and soil moisture, outrank static landscape features.

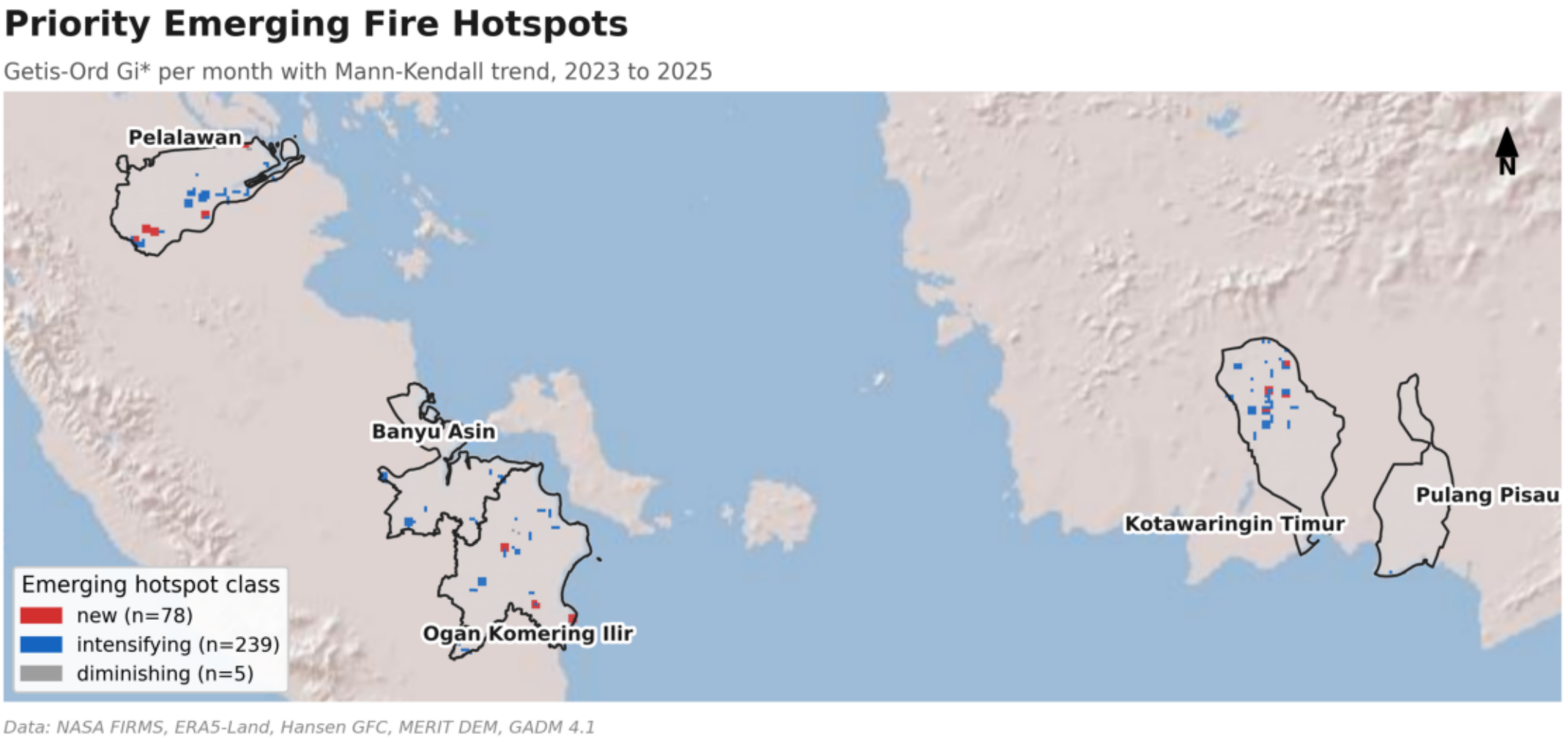
The Strongest Driver Interaction



NDWI: Normalized Difference Water Index | VPD: Vapour Pressure Deficit | LST: Land Surface Temperature

Compound dryness, dry air with dry soil, is the strongest signal. Warnings should watch both together.

Priority Emerging Hotspots



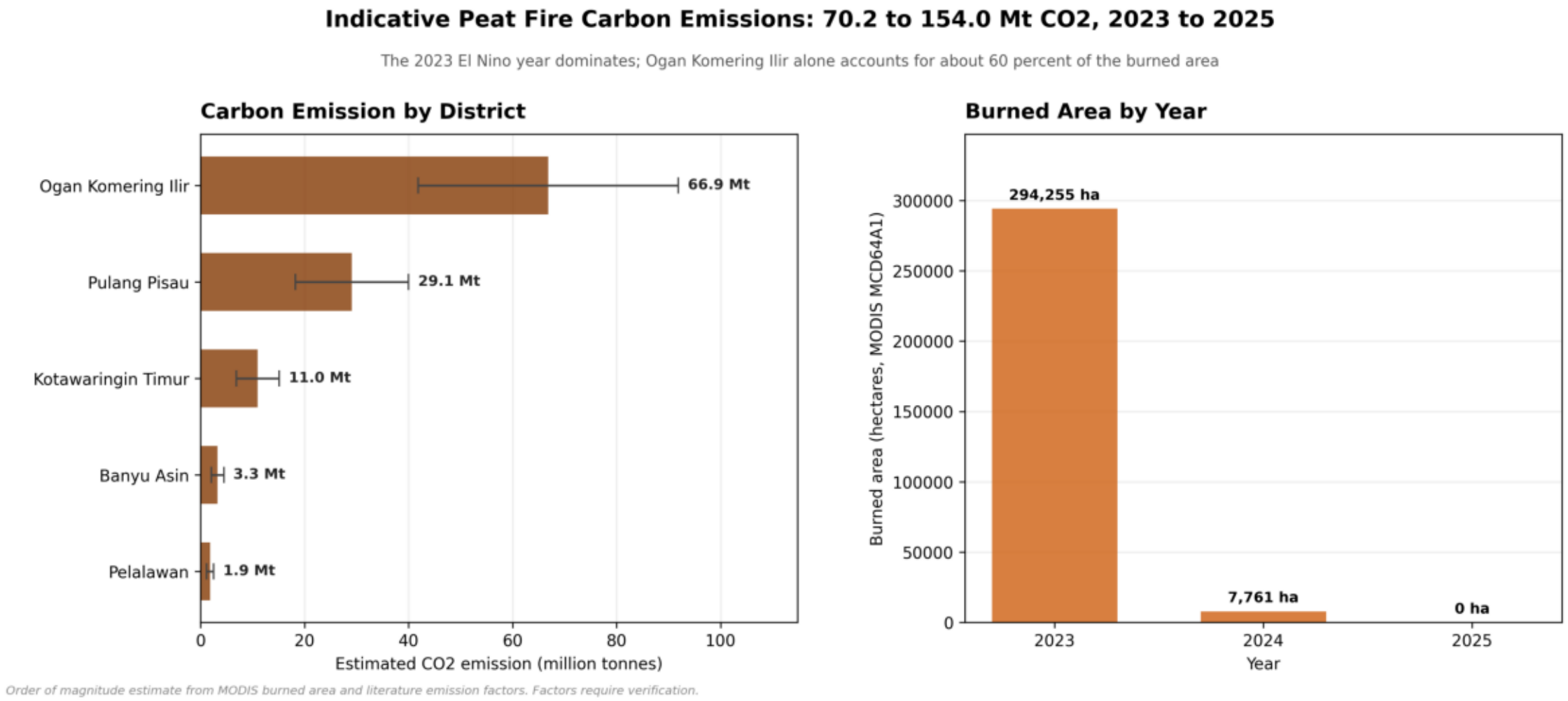
Blue cells are intensifying, red cells are newly active. These deserve priority monitoring.

Dominant Smoke Export Direction



On the worst days, winds carry smoke northwest toward the Malacca Strait.

Estimated Carbon Emissions



Indicative emissions of 70 to 154 million tonnes CO2, dominated by 2023. Emission factors require verification.

Recommendations

Three priorities for the next dry season

1

Move resources before the peak

Stage suppression crews and pumps in the highest risk district ahead of the dry season.

2

Warn on compound dryness

Trigger alerts when dry air and dry soil arrive together, not on a single indicator.

3

Flag haze days in advance

Coordinate transboundary warnings when heavy fire activity meets northwesterly winds.

Different districts need different responses. One policy does not fit all peatlands.

Limitations

Points to keep in mind when reading the results

- Peat extent is an elevation based proxy, not the licensed national peat map.
- Model skill is moderate, a screening tool rather than a precise forecaster.
- Smoke direction is a wind based estimate, not a physical dispersion model.
- Carbon estimate and the El Nino index values require verification.
- Pelalawan is hard to predict from climate alone, suggesting human driven ignition.